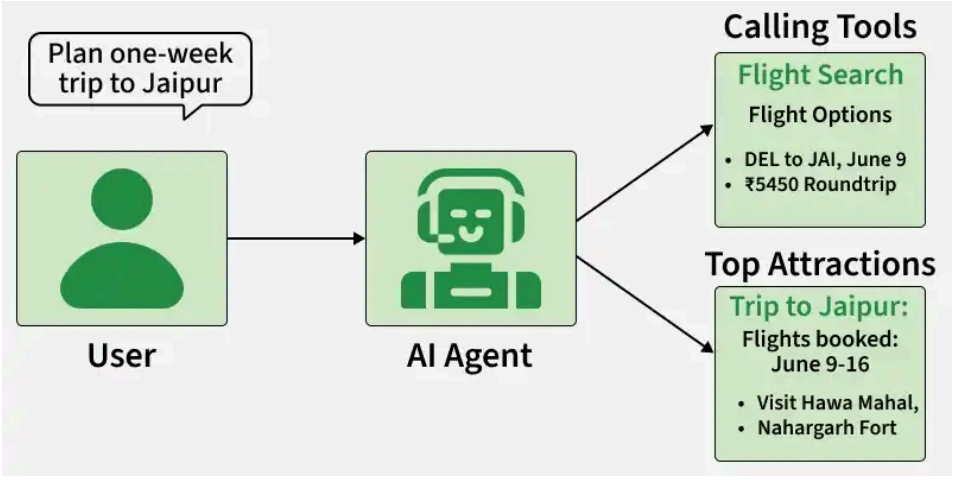
Agentic AI

Agentic AI is a type of AI system that can autonomously make decisions, plan actions and execute tasks to achieve specific goals with minimal human intervention. It focuses on goal-driven behavior, reasoning and interaction with tools and environments.

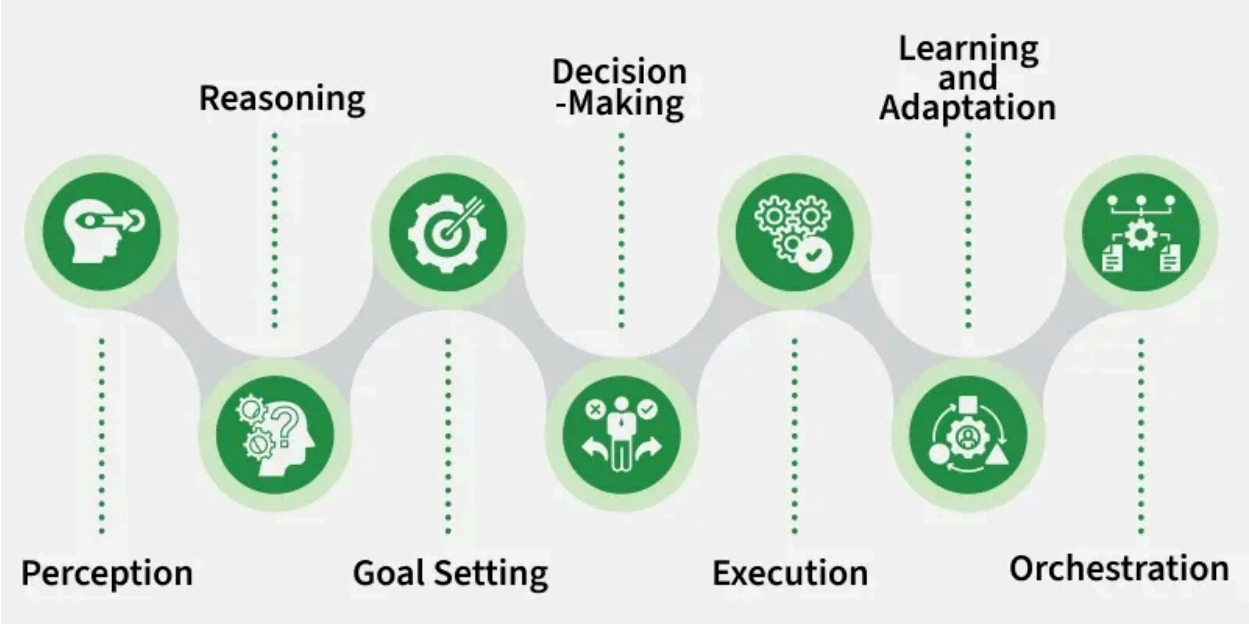
- **Specialised mastery:** It's trained and fine-tuned to handle a particular type of problem with great accuracy.
- **Tool usage:** It can connect with and use specific tools like software, APIs, databases, etc to achieve its goal.
- **Goal-oriented actions:** Instead of just giving information, it actively takes steps toward completing the task.
- **Efficient problem-solving:** Because it can plan, adapt and take actions autonomously, it can handle tasks more efficiently in dynamic environments.
- **Traditional vs. Agentic AI:** Unlike traditional AI systems that primarily respond to inputs, Agentic AI focuses on autonomous decision-making and goal-driven actions. It is defined by behavior (how it acts), not by the breadth of knowledge like General AI.



Key Characteristics of Agentic AI

- Autonomy and Goal-Oriented Behaviour: Agentic AI systems act independently and make decisions to achieve predefined goals without human intervention.
- Adaptive Learning and Complex Decision-Making: These systems learn from experience and adapt their behaviour to handle complex situations effectively.
- Environment Interaction and Perception: Agentic AI collects real-time data from its surroundings to understand and respond to the environment.
- Information Processing: It analyses data using algorithms and models to make informed decisions.
- Action Execution: The system performs tasks automatically using software commands or physical mechanisms based on its decisions.

How Agentic AI Works



Agentic AI systems operate through various steps such as:

- Perception: Collects relevant real-time data and retrieves past information from memory to provide context for the task.

- Reasoning: Interprets inputs using domain-specific knowledge and patterns stored in memory or knowledge bases.
- Goal Setting: Defines clear objectives and creates a focused plan based on input and past outcomes.
- Decision-Making: Chooses actions based on efficiency, accuracy and safety using past successful strategies.
- Execution: Performs tasks using tools like APIs and records results for future reference.
- Learning and Adaptation: Improves over time by learning from feedback and storing useful experiences.
- Orchestration: Collaborates with other agents in multi-agent systems to complete tasks efficiently.

Agentic AI vs. Traditional AI

Agentic AI and Traditional AI represent two different approaches used to build intelligent systems. Although often mentioned together, they are based on different principles and provide distinct capabilities. Let's see the key differences between them:

- Traditional AI requires human input and predefined rules, whereas Agentic AI operates independently and makes its own decisions.
- Traditional AI handles specific, routine tasks while Agentic AI adapts and learns to manage complex, dynamic goals.
- Decision-making in traditional AI is limited and rule-based but Agentic AI considers multiple variables and improves its strategies over time.
- Traditional AI struggles with changing environments, whereas Agentic AI continuously adapts to new information and evolving situations.

Feature	Traditional AI	Agentic AI
Core Function	Performs specific, preprogrammed tasks	Executes tasks autonomously using predefined goals
Typical output	Deterministic results—answers, classifications, predictions	Actions, decisions, multi-step workflows
Autonomy	Low as it requires explicit instructions, operates within set boundaries	High as it plans, adapts and makes decisions with minimal human direction
Learning	Learns from labeled data, often needs retraining for new situations	Learns from experience, adapts strategies and workflows in real time

Use cases	Data sorting, image recognition, basic diagnostics	Workflow automation, dynamic planning, virtual assistants, problem solving
Scalability	Requires manual oversight as systems grow.	Oversees and coordinates whole systems hence reducing manual monitoring.
Adaptability	Struggles with unexpected changes and may need retraining.	Adjusts strategies and learns in real time and is best suited for fast-changing situations.
Business value	Automates simple, rule-based jobs, increases consistency	Automates complex operations, reduces manual work, enables personalized tasks

Core capabilities of Agentic AI

1. Perception Module
What it does: Processes and interprets incoming data from the environment, such as text prompts, sensor feeds, images, or API responses.
Example: A chatbot reads a user's question, or a robot vacuum uses a camera to detect an obstacle in the room.

2. Reasoning & Decision-Making Module (The Brain)
What it does: Acts as the core intelligence (often powered by a Large Language Model or heuristic algorithms) that evaluates inputs, analyzes patterns, and determines the next best step.
Example: The agent weighs whether a customer request requires looking up a database or escalating to a human support agent.

3. Memory System
What it does: Retains and recalls information across short-term interactions (working memory) and long-term storage (such as vector databases) so the agent maintains context.
Example: A customer service agent remembers details the user mentioned earlier in the conversation thread.

4. Planning & Task Decomposition
What it does: Breaks down a broad, high-level goal into a sequence of smaller, manageable sub-tasks and adjusts the strategy if conditions change.
Example: Instead of just answering a question, the agent writes a multi-step plan to search the web, run a calculation, format a report, and email it.

5. Action & Tool Execution
What it does: Interfaces with external systems, software APIs, or physical actuators to carry out decisions and affect the environment.

Example: Calling a weather API, running a line of code, writing a file, or moving a robotic arm.

Single agent vs Multi agent system

Dimension	Single-agent system	Multi-agent system
Architecture	One LLM-driven agent, with or without tools, executing a task end to end	Two or more agents that communicate, hand off, or work in parallel toward a shared goal
Cost	Low. One reasoning chain, minimal redundant calls	High. Coordination messages, context duplication, and role overhead multiply LLM calls
Latency	Low to moderate. Sequential but direct	Higher. Agents wait on each other, verify each other, and sometimes loop
Reliability	One point of failure, but that point is easy to isolate	Errors from one agent can propagate to others before anyone catches them
Debuggability	Simple. One trace, one decision path	Hard. Failures hide inside inter-agent handoffs and shared state
Control	High. Deterministic, easy to constrain	Lower. Behavior emerges from agent interaction, harder to predict

Scalability	Scales by running more parallel instances of the same agent	Scales by adding roles, but each addition adds coordination surface
Best fit	Linear workflows, single-domain tasks, tool-augmented automation	Genuinely parallel work, distinct expert roles, tasks exceeding one context window

Use cases of Agentic AI :

Agriculture

In agriculture, AI agents help farmers increase yield while reducing waste.

The technology is capable of independently monitoring weather forecasts and soil conditions to optimize planting schedules and soil conditions. By continuously learning from environmental data and other inputs, AI agents help farmers make sustainable and cost-effective decisions to improve productivity.

Banking and financial services

In banking and financial services, agentic AI can be used to improve decision-making, optimize workflows and enhance compliance.

For instance, autonomous AI is used to perform continuous, autonomous fraud detection to detect unusual patterns and respond to emerging threats. Using similar logic, it’s well positioned to assist with compliance monitoring and loan underwriting, both of which involve a high volume of data-intensive repetitive tasks.

According to a recent World Economic Forum report, a suite of well-designed agentic AI pilots for financial institutions provided a raft of benefits. Together, they saved an estimated 30,000 workdays, and

improved productivity on administrative tasks, documentation and preliminary analysis by 20%-59%.

On the customer-facing side, AI agents and agentic virtual assistants provide AI-driven financial advisory services. For instance, they can automate certain wealth management activities or craft investment strategies based on market conditions and individual risk tolerance. By using AI solutions for financial management, enterprises mitigate potential disruptions and use data to maximize value and increase operational efficiency.

Disaster response
In disaster scenarios, AI agents can provide real-time intelligence and decision-making support for first responders.

These systems analyze satellite imagery, sensor networks and social media to assess damage and prioritize emergency response efforts. Predictive models and simulations also help localities prepare for future events. Tools such as these can enable proactive evacuations and minimize casualties, saving lives and reducing disaster response costs.

Education
In education, agentic AI tutors and learning platforms provide personalized and scalable learning paths for individual students.

AI-powered mentoring agents assess a student's level of knowledge and track their progress. They then adapt content in real time. This adaptive-learning capability helps ensure that all learners receive appropriately paced instruction.

These agents can independently generate exercises, give feedback and explain context when students struggle with certain concepts. They're also useful in responding to and learning from students' divergent learning styles.

In higher education, AI research assistants can help students explore topics by gathering sources or summarizing information.

In addition, language learning apps and career training platforms increasingly integrate autonomous agents that simulate real-world interactions such as job interviews or foreign-language conversations. These tailored experiences can lower the barriers to creating engaging simulations and give a greater number of students the opportunity to practice real-world skills.

Together, these tools have the potential to transform education into a more interactive, continuously evolving experience—resulting in increased student engagement and improved learning outcomes.

Healthcare
In healthcare, AI agents can provide more accurate diagnosis, highly personalized treatment plans and faster research-based innovations.

AI solutions have been of particular interest to the healthcare industry in recent years, given their ability to autonomously investigate health data. They also help remove administrative burdens in busy medical institutions. In clinical settings, AI agents given access to large datasets from across departments can significantly impact time spent on administrative tasks such as billing, scheduling and resource allocation. They can also completely automate routine tasks such as prior authorizations and remote patient monitoring.

Given their proactive approach to data analysis, AI agents can also help in diagnostics, manage drug processes and monitor patient vitals in real-time, flagging potential health risks before they escalate. By integrating agentic AI into everyday operations, hospitals and medical centers can make more informed decisions, allowing providers more time to focus on high-touch, personal care.